

# Food Hub Project

## Post Graduate Program in Artificial Intelligence and Machine Learning

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# Executive Summary



The analysis of FoodHub's order, operational, and customer rating data highlights clear trends in cuisine popularity, pricing behavior, operational efficiency, and revenue distribution. American and Japanese cuisines dominate the platform in both order volume and weekend demand, followed by Italian and Chinese cuisine types. Order frequency spikes significantly on weekends, with more than twice as many orders as weekdays.

Revenue is concentrated among a small set of high-performing restaurants. Four establishments—Shake Shack, The Meatball Shop, Blue Ribbon Sushi, and Blue Ribbon Fried Chicken—contribute roughly 30% of total gross revenue. Net revenue is estimated at \$6,166.

Operational performance appears consistently strong. Median preparation times (27 minutes) and delivery times (25 minutes) fall within predictable ranges, with distributions showing slight skewness but overall stability. Delivery times follow an approximately normal distribution centered between 20 and 30 minutes. Weekend operations achieve faster delivery speeds (median ~22 minutes) but with more variability, while weekdays show slower but more consistent delivery times (median ~28 minutes). Roughly 10% of orders exceed a 60-minute delivery time, highlighting an opportunity for targeted operational improvements.

Overall, FoodHub's data reveals strong weekend demand, concentrated revenue drivers, efficient operational performance, but significant gaps in rating completeness—over 700 unrated orders—which may distort customer satisfaction analysis. The insights point to opportunities to improve data completeness, evaluate outlier-prone cuisines, optimize slow or inconsistent delivery segments, and strengthen partnerships with top-performing restaurants.

# Conclusions & Recommendations

## Demand and Restaurant Strategy

- ❑ Prioritize and Support top-performing restaurants (Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken) since they generate ~30% of revenue.
- ❑ Expand partnership in American and Japanese cuisines, which show highest order volumes and weekend demand.
- ❑ Evaluate Underperforming cuisines (Korean, Vietnamese, Mediterranean) for menu optimization or promotional support.

## Pricing & Revenue Optimization

- ❑ Introduce targeted promotions for higher cost cuisines (Thai, French) to improve conversion while managing customer price sensitivity.
- ❑ Explore upsell or bundle options for moderately priced cuisines (~\$15 median) to lift average order value.
- ❑ Analyze outlier in cuisine pricing to ensure consistency and eliminate potential mispriced items.

## Operational Efficiency

- ❑ Investigate the 10% of orders with delivery times > 60 minutes to identify bottlenecks.
- ❑ Focus on stabilizing weekend delivery variability by leveraging demand forecasting.
- ❑ Review preparation time variability in Thai cuisine (highest variability) and replicate consistency practices from Korean cuisine (lowest variability).

# Conclusions & Recommendations

## Customer Ratings and Experience

- ☐ Improve rating capture by setting up a customer loyalty program to reward survey participation.
- ☐ Monitor ratings of '3' as they indicate the most operational variability and possible customer DSAT.
- ☐ Use qualitative feedback since ratings show little correlation with time or cost, suggesting perception-based drivers.

## Data & Analytical Improvements

- ☐ Investigate cuisine-specific outlines (cost and pre time) to identify data anomalies or true operational issues.
- ☐ Enhanced data completeness to strengthen reliability of future insights.
- ☐ Develop tracking and monitoring mechanism to monitor delivery consistency across weekdays and weekends.

# Business Problem Overview and Solution Approach

FoodHub has collected detailed data from customer orders placed through its mobile app platform. The company aims to analyze this data to gain insights into restaurant popularity, customer preferences, and operational performance. The analysis focuses on key variables, including cuisine type, restaurant name, order frequency, preparation time, delivery time, and customer ratings.

## Methodology:

1. Data Preprocessing
  - Review and clean the dataset by checking for missing values and duplicates.
  - Standardize data formats and ensure all variables have appropriate data types.
2. Exploratory Data Analysis (EDA)
  - **Univariate Analysis:** Examine individual variables to understand their distributions, patterns and characteristics.
  - **Multivariate Analysis:** Investigate relationships between two or more variables to identify correlations and patterns.
3. Statistical Analysis and Visualization
  - Apply statistical techniques to summarize key metrics.
  - Use visualizations such as histograms, boxplots and heatmaps to identify trends and relations.

The insights derived from the analysis will guide strategic decisions, including targeting specific customer segments, optimizing product offerings, improving operational efficiencies, and enhancing overall customer satisfaction.

# Data Overview 1

- Q1: The dataset used for this analysis is stored in an Excel file (foodhub\_order.xls) that includes 9 variables (columns) and 1,898 observations (rows).
- The data dictionary for this file is organized as follows:
  1. order\_id: Unique seven-digit numeric ID
  2. customer\_id: Unique numeric ID
  3. restaurant\_name: Restaurant Name
  4. cuisine\_type: Cuisine type ordered by the customer
  5. cost\_of\_the\_order: Cost of the Order
  6. day\_of\_the\_week: Day when the order was placed
  7. rating: Rating given by the customer
  8. food\_preparation\_time: Time taken by the restaurant to prepare the food (minutes)
  9. delivery\_time: Time taken to deliver the food order (minutes)

# Data Overview 2

- Q2: What are the datatypes of the different columns in the dataset?
  - Four variables have an integer data type.
  - Four variables have an object data type.
  - One variable have a float data type.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1898 entries, 0 to 1897
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   order_id              1898 non-null   int64
1   customer_id           1898 non-null   int64
2   restaurant_name       1898 non-null   object
3   cuisine_type          1898 non-null   object
4   cost_of_the_order     1898 non-null   float64
5   day_of_the_week       1898 non-null   object
6   rating                1898 non-null   object
7   food_preparation_time 1898 non-null   int64
8   delivery_time         1898 non-null   int64
dtypes: float64(1), int64(4), object(4)
memory usage: 133.6+ KB
```



# Data Overview 3

- Q3: Are there any missing values in the data?
  - There are no missing values in this dataset.
- Q4: Check the statistical summary of the data. What is the minimum, average, and maximum time it takes for food to be prepared once an order is placed?

|                                                                        |       |            |             |                   |                       |               |
|------------------------------------------------------------------------|-------|------------|-------------|-------------------|-----------------------|---------------|
| MIN: 20.00 minutes<br><br>AVG: 27.37 minutes<br><br>MAX: 35.00 minutes | ***   |            |             |                   |                       |               |
|                                                                        |       | order_id   | customer_id | cost_of_the_order | food_preparation_time | delivery_time |
|                                                                        | count | 1898.00    | 1898.00     | 1898.00           | 1898.00               | 1898.00       |
|                                                                        | mean  | 1477495.50 | 171168.48   | 16.50             | 27.37                 | 24.16         |
|                                                                        | std   | 548.05     | 113698.14   | 7.48              | 4.63                  | 4.97          |
|                                                                        | min   | 1476547.00 | 1311.00     | 4.47              | 20.00                 | 15.00         |
|                                                                        | 25%   | 1477021.25 | 77787.75    | 12.08             | 23.00                 | 20.00         |
|                                                                        | 50%   | 1477495.50 | 128600.00   | 14.14             | 27.00                 | 25.00         |
|                                                                        | 75%   | 1477969.75 | 270525.00   | 22.30             | 31.00                 | 28.00         |
|                                                                        | max   | 1478444.00 | 405334.00   | 35.41             | 35.00                 | 33.00         |

- Q5: How many orders are not rated?
  - 736 orders don't have a customer rating (~38% of the total orders).

# Univariate Analysis 1

A Univariate analysis was conducted for all nine variables in the dataset to understand their individual distributions, patterns, and characteristics. The techniques used in this analysis included:

1. **Summary Statistics** – to evaluate key measures such as mean, median, mode, range and standard deviation.
2. **Frequency Distributions** – to observe the occurrences of categorical values and identify common or dominant categories.
3. **Other Statistical Techniques** – such as visual inspections using histograms, boxplots and count plots to detect skewness, outliers, and overall data spread.

| Variable           | order_id    | customer_id | restaurant_name | cuisine_type | cost_of_the_order     | day_of_the_week | rating                   | food_preparation_time | delivery_time         |
|--------------------|-------------|-------------|-----------------|--------------|-----------------------|-----------------|--------------------------|-----------------------|-----------------------|
| Type               | Categorical | Categorical | Categorical     | Categorical  | Numeric               | Categorical     | Numeric                  | Numeric               | Numeric               |
| Analysis           | Uniqueness  | Uniqueness  | Freq counts     | Freq counts  | Histograms & Boxplots | Freq counts     | Uniqueness & Freq counts | Histograms & Boxplots | Histograms & Boxplots |
| Summary Statistics | Yes         | Yes         | N/A             | N/A          | Yes                   | N/A             | N/A                      | Yes                   | Yes                   |

# Univariate Analysis 2

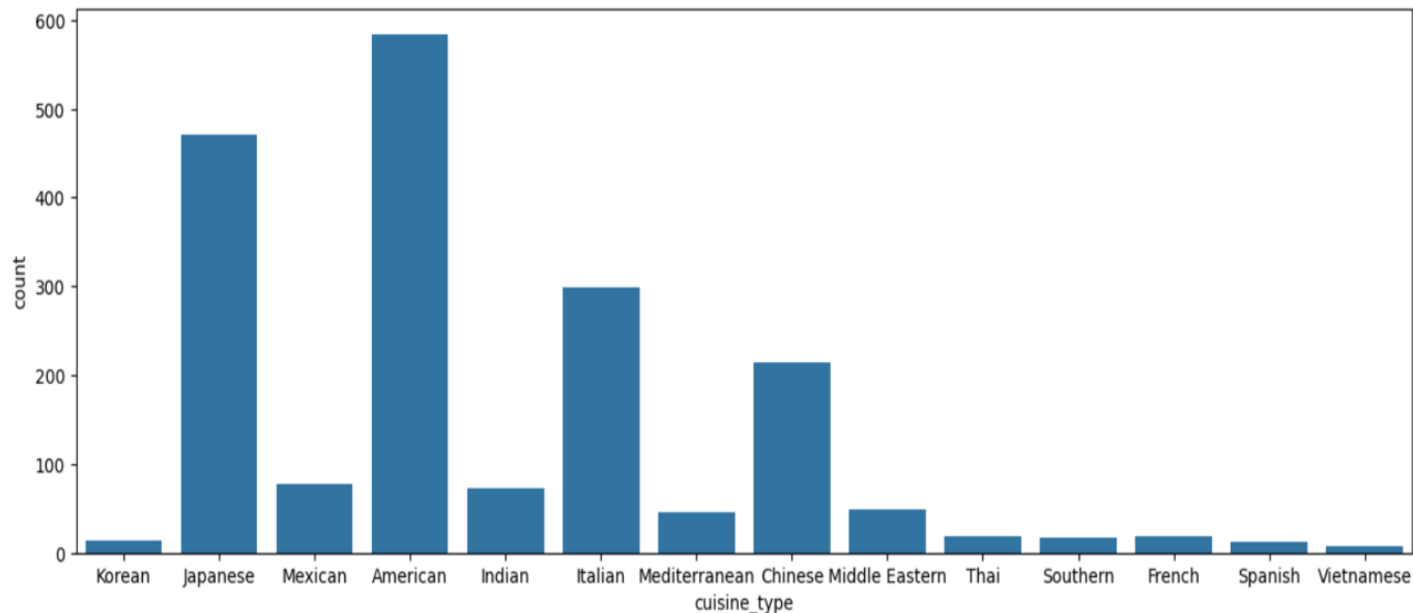
- Q6: Explore all the variables and provide observations on their distributions. (Generally, histograms, boxplots, countplots, etc. are used for univariate exploration.)
  - a) order\_id: This variable contains 1898 unique values.
  - b) customer\_id: This variable contains 1200 unique values.
  - c) restaurant\_name: This variable contains 178 unique values. Some examples include: 'Hangawi', 'Blue Ribbon Sushi Izakaya', 'Cafe Habana', 'Blue Ribbon Fried Chicken', 'Dirty Bird to Go'.

```
... array(['Hangawi', 'Blue Ribbon Sushi Izakaya', 'Cafe Habana',  
         'Blue Ribbon Fried Chicken', 'Dirty Bird to Go',  
         'Tamarind TriBeCa', 'The Meatball Shop', 'Barbounia',  
         'Anjappar Chettinad', 'Bukhara Grill',  
         'Big Wong Restaurant \x8c\x94\x94', 'Empanada Mama (closed)', 'Pylos',  
         "Lucky's Famous Burgers", 'Shake Shack', 'Sushi of Gari',  
         'RedFarm Hudson', 'Blue Ribbon Sushi',  
         'Five Guys Burgers and Fries', 'Tortaria', 'Cafe Mogador',
```

# Univariate Analysis 3

d) cuisine\_type: This variable contains 14 unique values

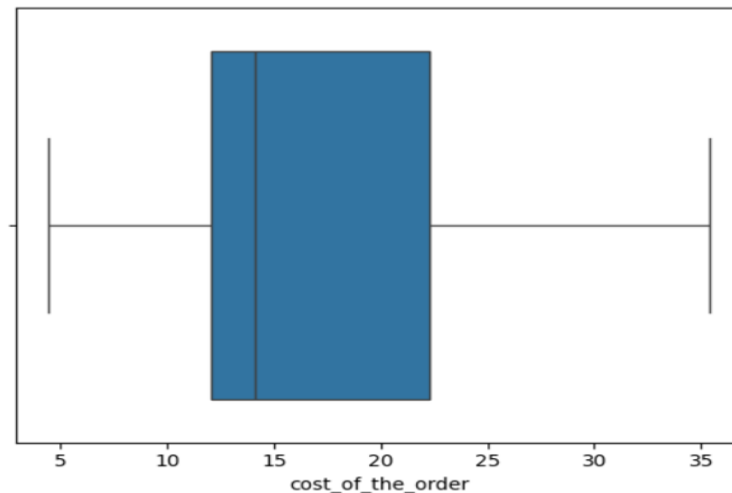
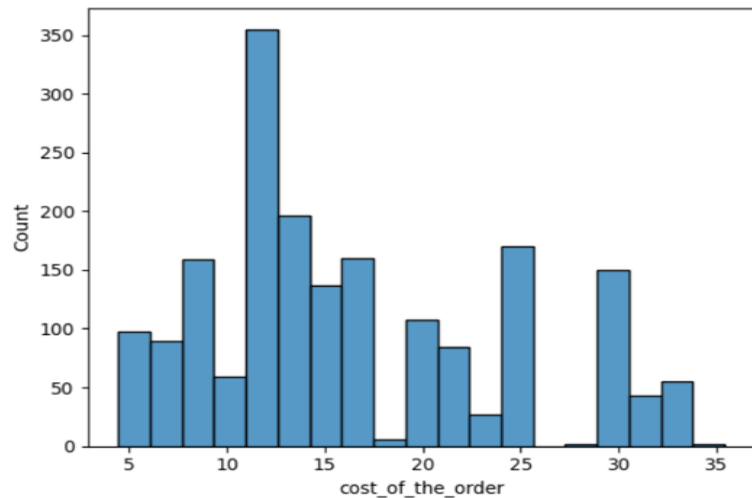
- Among all cuisine types, American and Japanese restaurants show the largest representation, with Italian and Chinese restaurants ranking next in frequency.



# Univariate Analysis 4

e) cost\_of\_the\_order:

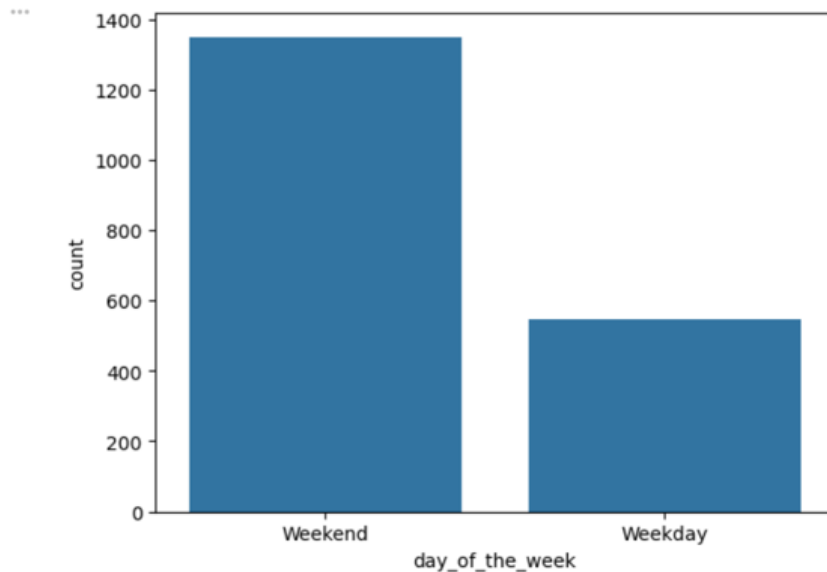
- 50% of the cost order values are less than \$15.
- Cost of the order does not have any outliers.
- The distribution of cost of order is highly right skewed - this indicates that most orders are moderately priced, but some can be quite expensive.



# Univariate Analysis 5

f) day\_of\_the\_week:

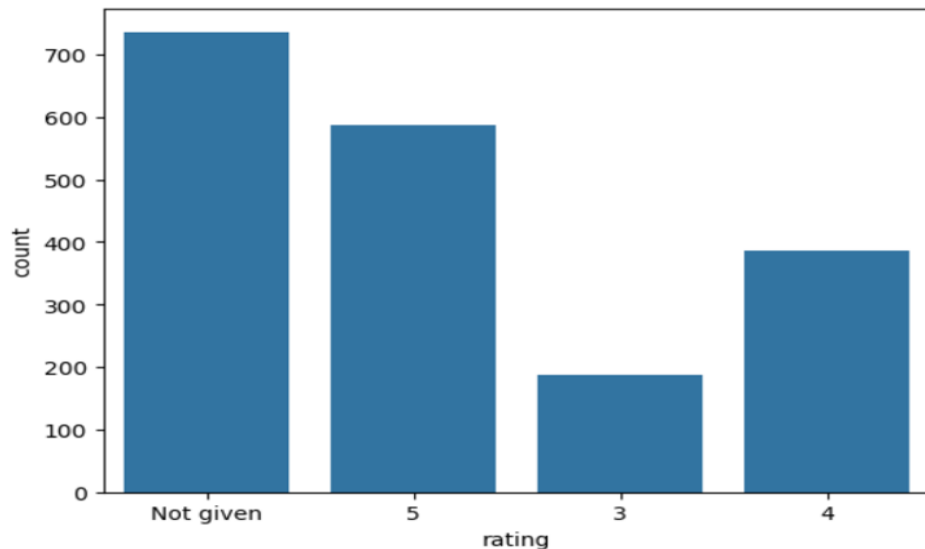
- The number of orders is more than double on weekends than on weekdays.



# Univariate Analysis 6

g) rating:

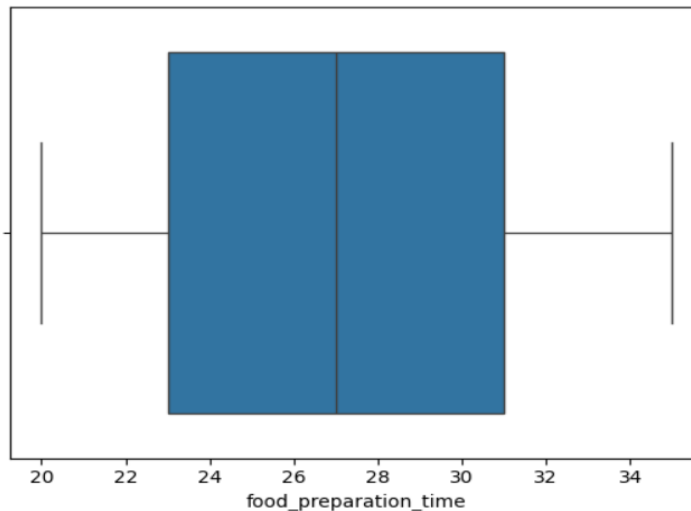
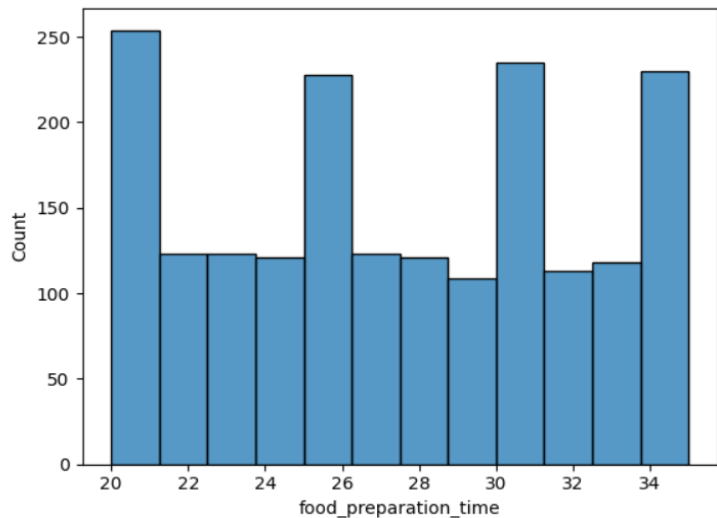
- Significant portion of rating counts (700) lack a rating.
- Satisfaction levels appear high (5) but the absence of rating counts may skew the analysis.



# Univariate Analysis 7

h) food\_preparation\_time

- 50% of the orders are prepared in less than 27 minutes.
- The distribution of the food preparation time histogram is bimodal with two peaks at 20 minutes and 30 minutes.
- The distribution of preparation times shows a slight positive skew, indicating that approximately 25% of orders (the 75th percentile) have preparation times ranging from 27 to 31 minutes.

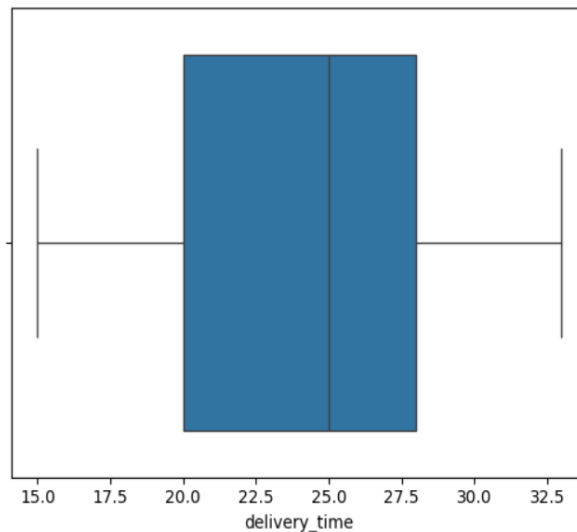
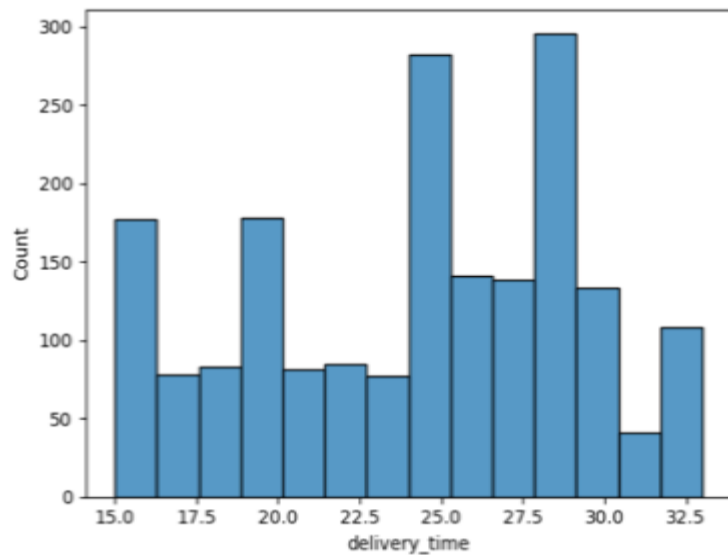




# Univariate Analysis 8

i) delivery\_time

- 50% of the orders are delivered in less than 25 minutes.
- The delivery time histogram shows an approximately normal distribution, with most orders concentrated around 20 to 30 minutes.



# Univariate Analysis 9

- Q7: Which are the top 5 restaurants in terms of the number of orders received?
  - The top five restaurants with the highest number of orders are Shake Shack, The Meatball Shop, Blue Ribbon Sushi, Blue Ribbon Fried Chicken, and Parm.

| restaurant_name           | count |
|---------------------------|-------|
| Shake Shack               | 219   |
| The Meatball Shop         | 132   |
| Blue Ribbon Sushi         | 119   |
| Blue Ribbon Fried Chicken | 96    |
| Parm                      | 68    |

# Univariate Analysis 10

- Q8: Which is the most popular cuisine on weekends?
  - The most popular weekend cuisine is American with 415 orders, followed by Japanese with 335.

| cuisine_type | count |
|--------------|-------|
| American     | 415   |
| Japanese     | 335   |
| Italian      | 207   |
| Chinese      | 163   |
| Mexican      | 53    |

# Univariate Analysis 11

- Q9: What percentage of the orders cost more than 20 dollars?
  - The number of total orders that cost above 20 dollars is: 555.
  - Percentage of orders above 20 dollars: 29.24%.
- Q10: What is the mean order delivery time?
  - The mean delivery time for this dataset is 24.16 minutes.
- Q11: The company has decided to give 20% discount vouchers to the top 5 most frequent customers. Find the IDs of these customers and the number of orders they placed

| count       |    |
|-------------|----|
| customer_id |    |
| 52832       | 13 |
| 47440       | 10 |
| 83287       | 9  |
| 250494      | 8  |
| 259341      | 7  |

# Multivariate Analysis 1



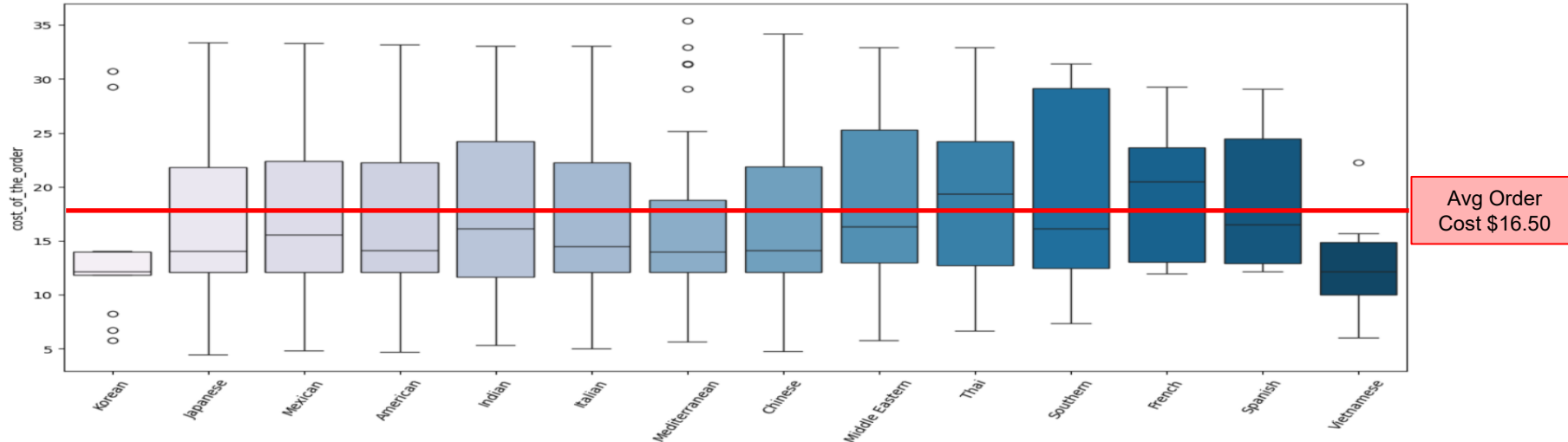
A Multivariate analysis was conducted to examine the relationships between two or more variables in the dataset to identify their patterns and correlations. The techniques used in this analysis included:

1. **Correlation Analysis** – to measure the strength of relationships.
2. **Heatmaps** – to visualize the relationships and patterns across multiple variables.
3. **Boxplots** – to compare measures of central tendency across multiple variables.

| Relation | Cost of the Order and Cuisine Type | Food Preparation Time and Cuisine Type | Day of the Week and Delivery Time | Rating and Delivery Time | Rating and Food Preparation Time | Rating and Cost of the Order | Cost of the Order, Food Preparation Time, and Delivery Time |
|----------|------------------------------------|----------------------------------------|-----------------------------------|--------------------------|----------------------------------|------------------------------|-------------------------------------------------------------|
| X        | Cost of the Order (Numeric)        | Cuisine Type (Categorical)             | Day of the Week (Categorical)     | Rating (Numeric)         | Rating (Numeric)                 | Rating (Numeric)             | All (Numeric)                                               |
| Y        | Cuisine Type (Categorical)         | Food Preparation Type (Numeric)        | Delivery Time (Numeric)           | Delivery Time (Numeric)  | Food Preparation Time (Numeric)  | Cost of the Order (Numeric)  | All (Numeric)                                               |
| Analysis | Boxplot                            | Boxplot                                | Boxplot                           | Pointplot                | Pointplot                        | Pointplot                    | Heatmap & Correlation                                       |

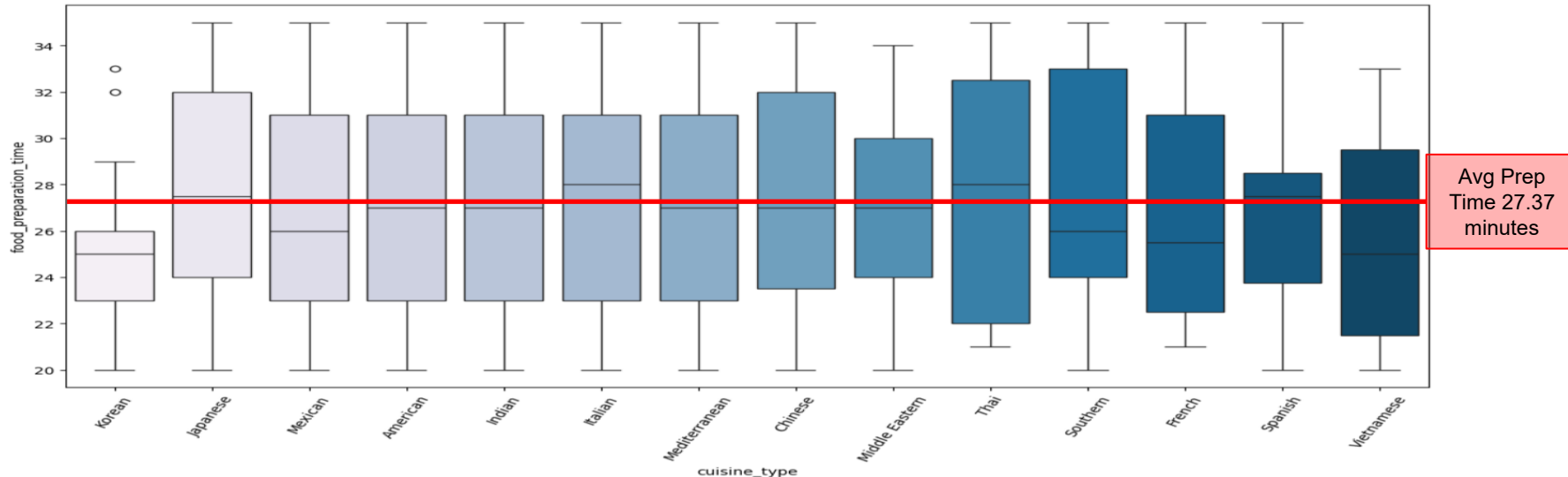
# Multivariate Analysis 2

- Q12: Relation between Cost of the Order and Cuisine Type
  - Thai and French cuisine have the highest median cost at ~\$19 -20 which could indicate a left skewed distribution considering the average cost is \$16.50 (Median > Avg).
  - American and Japanese cuisine (most order counts) have very similar boxplots with a median cost at ~\$15.
  - Korean and Vietnamese cuisine(lowest order counts) have the lowest median cost at ~\$12.
  - Korean, Mediterranean and Vietnamese have the presence of outliers that should be further analyzed.



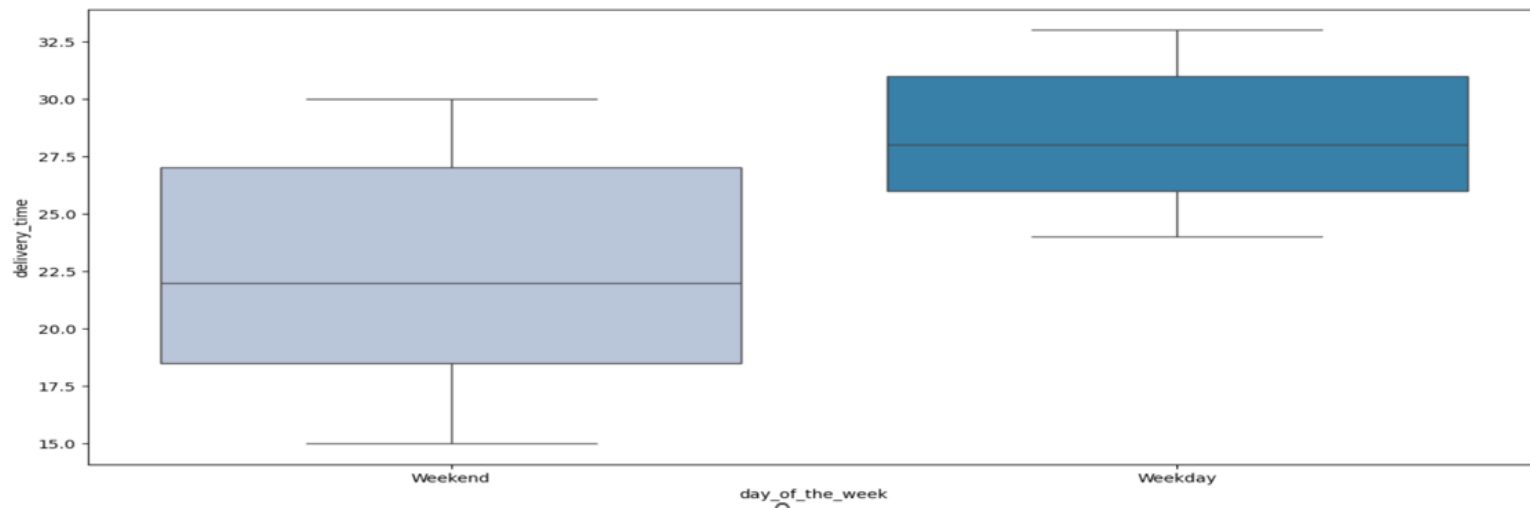
# Multivariate Analysis 3

- Q12: Relation between Food Preparation Time and Cuisine Type
  - Korean, French and Vietnamese cuisine have the lowest median preparation time at ~25 minutes which could indicate a right skewed distribution considering the average preparation time is 27.37 minutes (Median<Avg).
  - American and Japanese cuisine (most order counts) have very similar boxplots with a median preparation time at ~27 minutes – Both cuisine types show moderate prep time variability.
  - Korean cuisine shows the least prep time variability while Thai cuisine shows the most prep time variability.



# Multivariate Analysis 4

- Q12: Relation between Day of the Week and Delivery Time
  - Weekend median delivery time has the lowest delivery time at ~22 minutes with a wide-spread which indicates high variability and less consistency.
  - Weekday median delivery times has the highest delivery time at ~28 minutes with a narrow-spread which indicates low variability and more consistency.





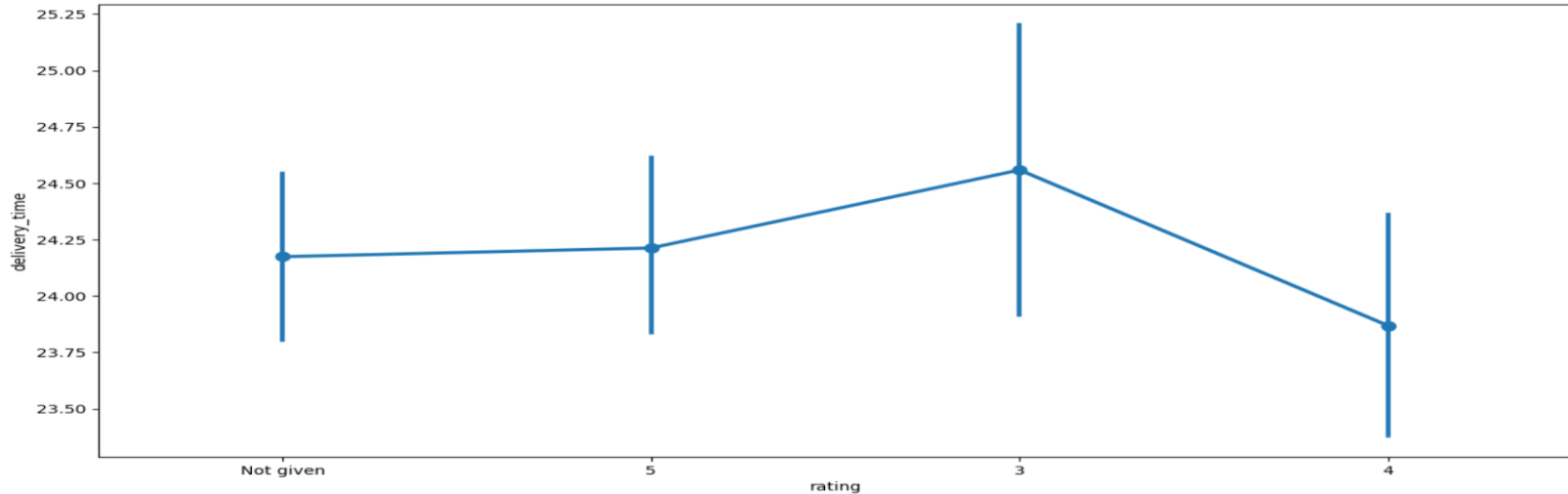
# Multivariate Analysis 5

- Q12: Revenue by Restaurants
  - Total gross revenue: \$31,314.
  - Four restaurants (Shake Shack; The Meatball Shop; Blue Ribbon Sushi and Blue Ribbon Fried Chicken) make ~30% of the total gross revenue.
  - Two American cuisine restaurant (Shake Shack & Blue Ribbon Fried Chicken) make 16% of the total gross revenue.

| restaurant_name               | cost_of_the_order |
|-------------------------------|-------------------|
| Shake Shack                   | 3579.53           |
| The Meatball Shop             | 2145.21           |
| Blue Ribbon Sushi             | 1903.95           |
| Blue Ribbon Fried Chicken     | 1662.29           |
| Parm                          | 1112.76           |
| RedFarm Broadway              | 965.13            |
| RedFarm Hudson                | 921.21            |
| TAO                           | 834.50            |
| Han Dynasty                   | 755.29            |
| Blue Ribbon Sushi Bar & Grill | 666.62            |
| Rubirosa                      | 660.45            |
| Sushi of Gari 46              | 640.87            |
| Nobu Next Door                | 623.67            |
| Five Guys Burgers and Fries   | 506.47            |

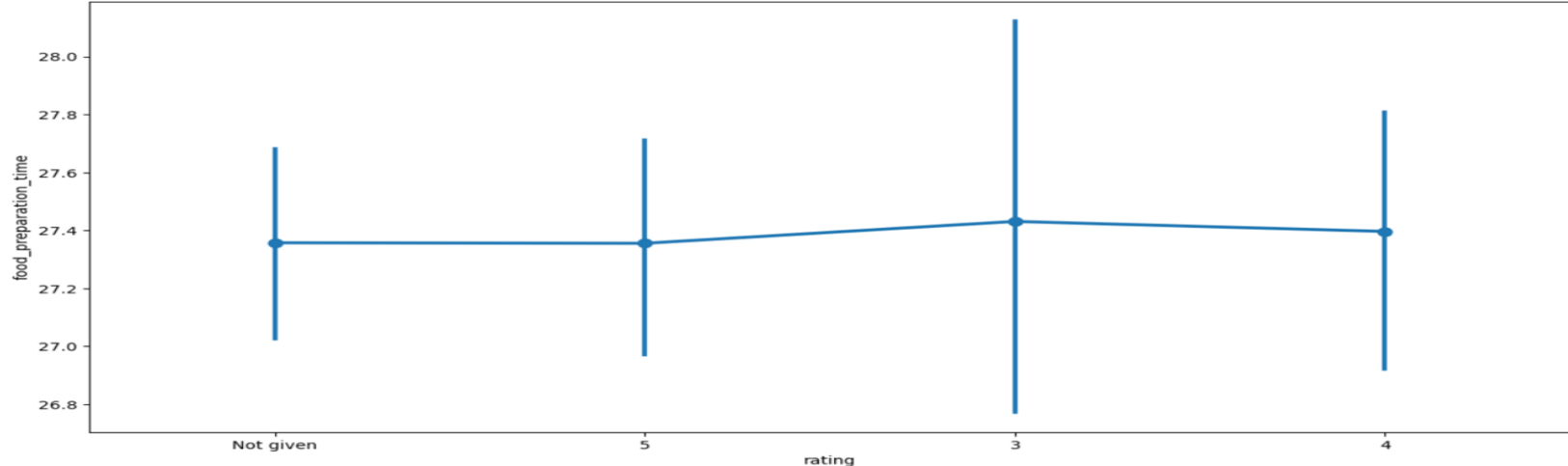
# Multivariate Analysis 6

- Q12: Relation between Rating and Delivery Time
  - Rating '3' has the longest average delivery time and the widest spread suggesting high variability and less consistency.
  - Rating '5' and 'Not Given' show similar spreads with an almost identical average delivery time.
  - Rating '4' shows the lowest average delivery time which indicates customers propensity to reward short delivery times.



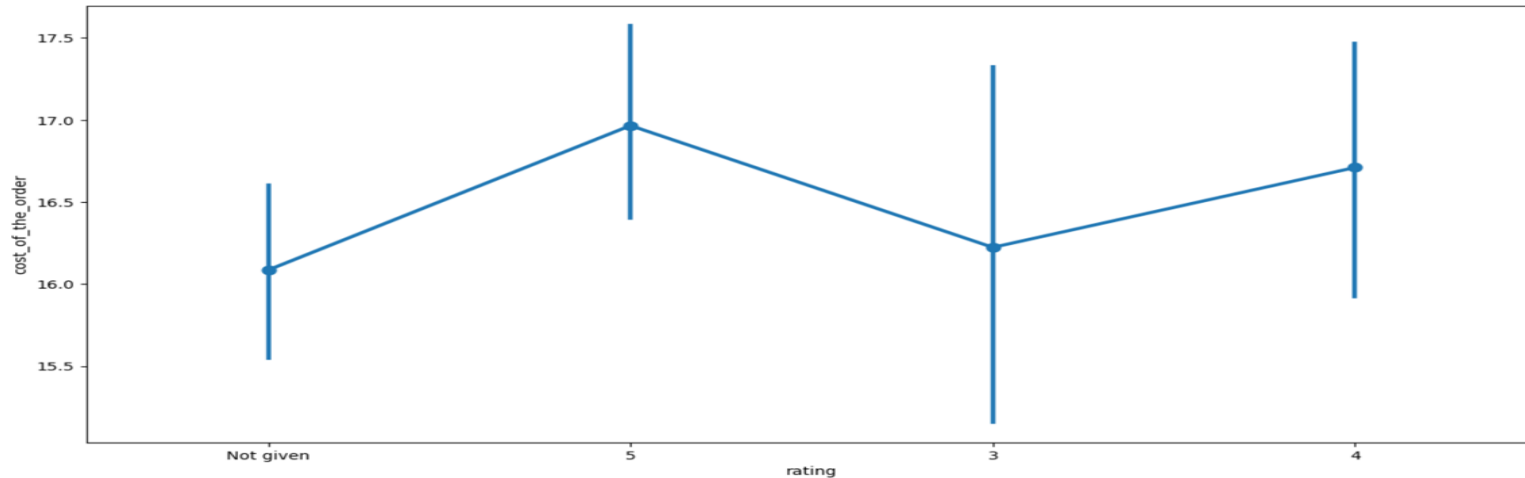
# Multivariate Analysis 7

- Q12: Relation between Rating and Food Preparation Time
  - Rating '3' has the longest food preparation time and the widest spread suggesting high variability and less consistency.
  - Rating '5' and 'Not Given' show similar spreads with an almost identical food preparation time.
  - The average food preparation time is consistent across all rating categories which indicates no correlation between rating and food preparation time



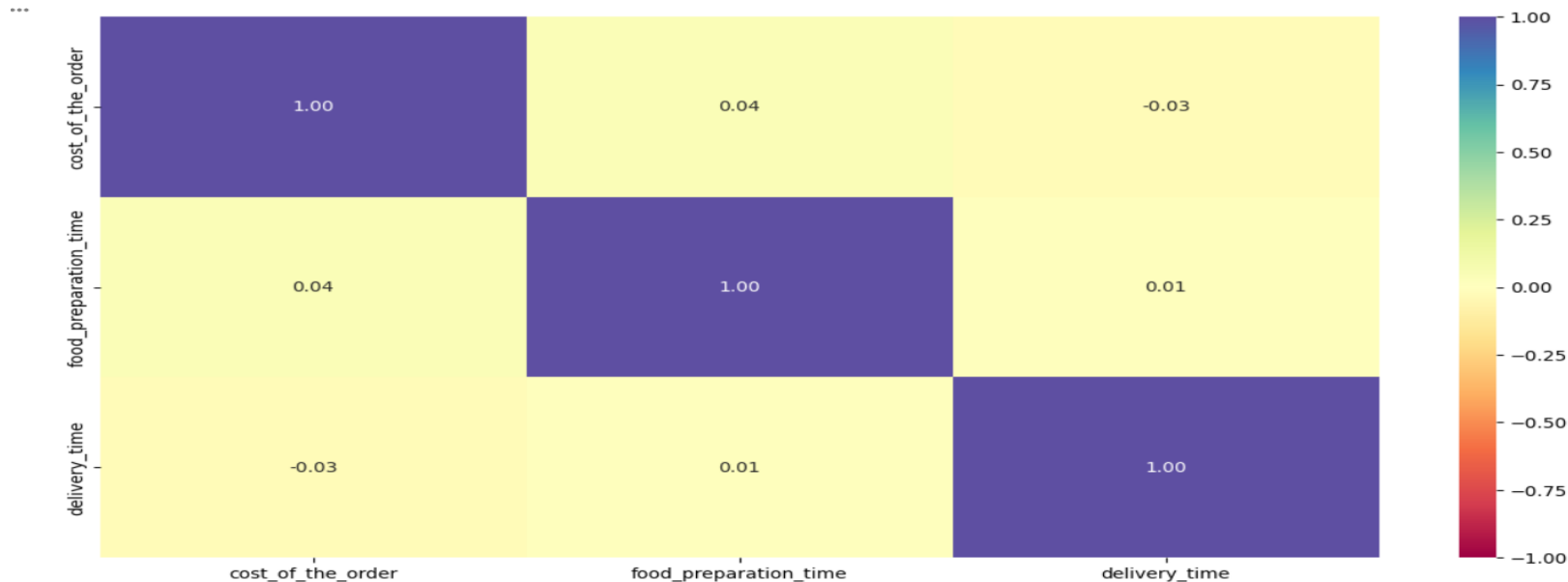
# Multivariate Analysis 8

- Q12: Relation between Rating and Cost of the Order
  - Rating '3' has the longest order cost range and the widest spread suggesting high variability and less consistency.
  - Rating '5' is associated with the highest order cost which indicates no correlation between cost and rating – value perception matters more than price alone.



# Multivariate Analysis 9

- Q12: Correlation among Cost of the Order, Food Preparation Time, and Delivery Time
  - There is no meaningful correlation across any combinations of these 3 variables
  - Metrics are largely uncorrelated which indicates that optimizing on variable will not cause any effect on the other variable



# Multivariate Analysis 10

- Q13: Promotional Offer: restaurants with a rating count of 50 and an average rating greater than 4
  - Four restaurants (The Meatball Shop, Shake Shack, Blue Ribbon Sushi and Blue Ribbon Fried Chicken) have a rating of 4 and a rating count greater than 50.

|   | restaurant_name           | rating |
|---|---------------------------|--------|
| 0 | Shake Shack               | 133    |
| 1 | The Meatball Shop         | 84     |
| 2 | Blue Ribbon Sushi         | 73     |
| 3 | Blue Ribbon Fried Chicken | 64     |
| 4 | RedFarm Broadway          | 41     |

|   | restaurant_name           | rating   |
|---|---------------------------|----------|
| 0 | The Meatball Shop         | 4.511905 |
| 1 | Shake Shack               | 4.278195 |
| 2 | Blue Ribbon Sushi         | 4.219178 |
| 3 | Blue Ribbon Fried Chicken | 4.328125 |

# Multivariate Analysis 11



- Q14: Company Net Revenue: Net revenue generated by the company across all orders
  - The Net Revenue is around 6166.3 dollars.

|   | order_id | customer_id | restaurant_name           | cuisine_type | cost_of_the_order | day_of_the_week | rating    | food_preparation_time | delivery_time | Revenue |
|---|----------|-------------|---------------------------|--------------|-------------------|-----------------|-----------|-----------------------|---------------|---------|
| 0 | 1477147  | 337525      | Hangawi                   | Korean       | 30.75             | Weekend         | Not given | 25                    | 20            | 7.6875  |
| 1 | 1477685  | 358141      | Blue Ribbon Sushi Izakaya | Japanese     | 12.08             | Weekend         | Not given | 25                    | 23            | 1.8120  |
| 2 | 1477070  | 66393       | Cafe Habana               | Mexican      | 12.23             | Weekday         | 5         | 23                    | 28            | 1.8345  |
| 3 | 1477334  | 106968      | Blue Ribbon Fried Chicken | American     | 29.20             | Weekend         | 3         | 25                    | 15            | 7.3000  |
| 4 | 1478249  | 76942       | Dirty Bird to Go          | American     | 11.59             | Weekday         | 4         | 25                    | 24            | 1.7385  |

# Multivariate Analysis 12

- Q15: Total Time Required to Deliver: % of orders that take more than 60 minutes to get delivered from the time the order is placed (preparation time + delivery time)
  - The total number of orders with a delivery time greater than 60 minutes: 200.
  - Percentage of orders with a delivery time above 60 minutes: 10.54%.
- Q16: Mean Delivery Times: how does the mean delivery time vary during weekdays and weekends?
  - The mean delivery time on weekdays is around 28 minutes.
  - The mean delivery time on weekends is around 22 minutes.
- Q17: What are your conclusions from the analysis? What recommendations would you like to share to help improve the business?
  - Refer to slides 3, 4 & 5.

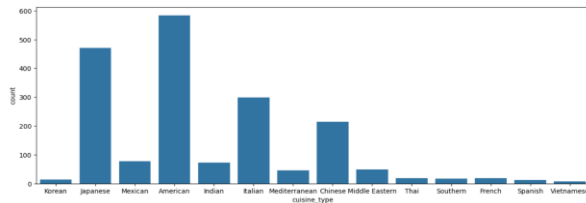


# APPENDIX

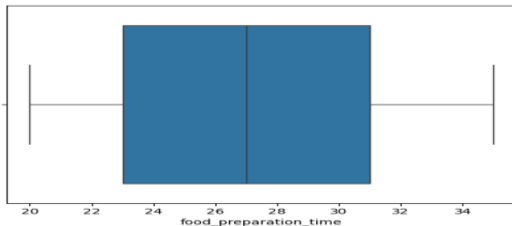
# Sample Code 1

```
""
order_id customer_id cost_of_the_order food_preparation_time delivery_time
count 1898.00 1898.00 1898.00 1898.00 1898.00
mean 1477495.50 171168.48 16.50 27.37 24.16
std 548.05 113698.14 7.48 4.63 4.97
min 1476547.00 1311.00 4.47 20.00 15.00
25% 1477021.25 77787.75 12.08 23.00 20.00
50% 1477495.50 128600.00 14.14 27.00 25.00
75% 1477969.75 270525.00 22.30 31.00 28.00
max 1478444.00 405334.00 35.41 35.00 33.00
```

```
df.describe().round(2)
```



```
plt.figure(figsize = (15,5))
sns.countplot(data = df, x = 'cuisine_type');
```



```
sns.boxplot(data=df,x='food_preparation_time')
plt.show()
```

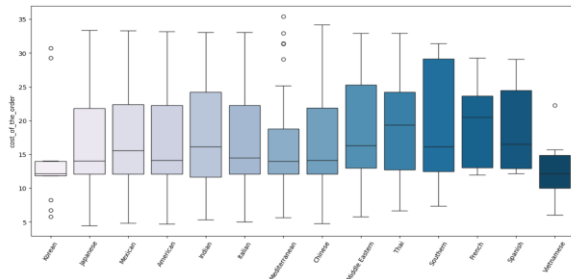
# Sample Code 2

The number of total orders that cost above 20 dollars is: 555

Percentage of orders above 20 dollars:  
29.24 %

```
df_greater_than_20 = df[df['cost_of_the_order']>20]
print('The number of total orders that cost above 20 dollars is:',
df_greater_than_20.shape[0])
```

```
percentage = (df_greater_than_20.shape[0] / df.shape[0]) * 100
print("Percentage of orders above 20 dollars:", round(percentage, 2), '%')
```

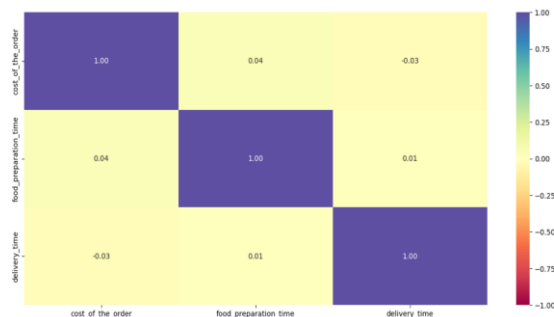


```
plt.figure(figsize=(15,7))
sns.boxplot(x = "cuisine_type", y = "cost_of_the_order", data = df, palette = 'PuBu', hue
= "cuisine_type")
plt.xticks(rotation = 60)
plt.show()
```

Total order cost: 31314.82

```
total_order_cost = df['cost_of_the_order'].sum()
print("Total order cost:", total_order_cost)
```

# Sample Code 3



```
col_list = ['cost_of_the_order', 'food_preparation_time', 'delivery_time']  
plt.figure(figsize=(15, 7))  
sns.heatmap(df[col_list].corr(), annot=True, vmin=-1, vmax=1, fmt=".2f",  
cmap="Spectral")  
plt.show()
```

```
" order_id customer_id restaurant_name cuisine_type cost_of_the_order day_of_the_week rating food_preparation_time delivery_time Revenue  
0 1477147 337325 Hangawi Korean 30.75 Weekend Not given 25 20 7.6875  
1 1477885 358141 Blue-Ribbon-Sushi Izakaya Japanese 12.08 Weekend Not given 25 23 1.8120  
2 1477070 66383 Cafe Habana Mexican 12.23 Weekday 5 23 28 1.8345  
3 1477334 106668 Blue-Ribbon-Fried-Chicken American 29.20 Weekend 3 25 15 7.3000  
4 1478249 76942 Ding-Ding-to-Go American 11.58 Weekday 4 25 24 1.7385
```

```
def compute_rev(x):
```

```
    if x > 20:  
        return x*0.25  
    elif x > 5:  
        return x*0.15  
    else:  
        return x*0
```

```
df['Revenue'] = df['cost_of_the_order'].apply(compute_rev)  
df.head()
```

# Sample Code 4



|                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>The mean delivery time on weekdays is around 28 minutes</p> <p>The mean delivery time on weekends is around 22 minutes</p>                                  | <pre>print('The mean delivery time on weekdays is around',<br/>      round(df[df['day_of_the_week'] == 'Weekday']['delivery_time'].mean()),<br/>      'minutes')<br/><br/>print('The mean delivery time on weekends is around',<br/>      round(df[df['day_of_the_week'] == 'Weekend']['delivery_time'].mean()),<br/>      'minutes')</pre>                                                            |
| <p>The number of total orders with a delivery time greater than 60 minutes: 200</p> <p>Percentage of orders with a delivery time above 60 minutes: 10.54 %</p> | <pre>df['total_time'] = df['food_preparation_time'] + df['delivery_time']<br/><br/>print('The number of total orders with a delivery time greater than 60 minutes:',<br/>      df_greater_than_60.shape[0])<br/><br/>percentage = (df_greater_than_60.shape[0] / df.shape[0]) * 100<br/><br/>print("Percentage of orders with a delivery time above 60 minutes:", round(percentage,<br/>2), '%')</pre> |